

class Test

1) An image appears too bright.

Grey level Transformation - Image Negative / Intensity Reduction

- => A bright image has high intensity values that reduce visibility
- => The most suitable grey level transformation is image negative to decrease excessive brightness
- => It transforms pixel values using  $s = L - 1 - r$ , where  $L$  is the maximum
- => This reduces the brightness and improves the visibility of image
- => Outcome: The image becomes balanced, and hidden features are easier to observe

2) A dark image lacks detail.

Log Transformation.

- => A dark image contains low-intensity pixels that hide important details.
- => Log transformation dark regions while preserving brighter area
- => Formula:  $s = c \log(I + \gamma)$

=> The brightens dark regions while preserving brighter area.

=> Outcome: Hidden details become visible, the overall image appears clearer.

3) An image has poor contrast.

### Histogram Equalization

=> Histogram equalization is used to improve image contrast.

=> It redistributes the pixel intensity values over the full gray-level range.

=> This increases the difference between dark and bright regions.

=> Outcome: The image becomes sharper with improved contrast and better visual quality.

4) A low-contrast image requires uniform intensity distribution.

### Histogram Equalization

=> Histogram Equalization spreads pixel intensities uniformly across the available range.



- ⇒ It calculates the cumulative histogram and maps old intensity values to new one.
- ⇒ This improves the overall contrast of the image
- ⇒ Outcome: Intensity values become more evenly distributed, making objects easier to identify.

An image shown uneven brightness across regions.

### Adaptive Histogram Equalization (CLAHE)

- ⇒ Uneven brightness can be corrected using contrast limited Adaptive Histogram Equalization
- ⇒ CLAHE divides the image into small regions and enhances each region separately.
- ⇒ It also limits contrast amplification to avoid noise.
- ⇒ Outcome: Brightness becomes uniform across the image, and local details are enhanced without over-amplifying noise.